EXATHON INDIA 2026 · THE EXASOL OPEN BUILD CHALLENGE

LoopCraft

Closed-Loop AI Support Optimization Engine

Transforming discarded customer negative feedback into verified, auto-updated knowledge base articles in real time.

⚡ Powered by Exasol Personal In-Memory Columnar Database

Track 04: The Work Behind the Work · Track 02: Customers, Channels & Demand

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Closed-Loop AI & KB Optimizer

🔄 Autonomous Deflection Learning

Intercepts thumbs-down signals and streams failure telemetry into in-memory columnar analytics in real time.

🔍 Stale Article Detection

Columnar window functions surface outdated KB articles causing repeated escalations across 50k+ conversations.

✅ 1-Click Admin Approvals

Auto-drafted correction briefs with voice audio summary ready for instant publishing with a single admin approval.

The Problem: The Support Feedback Black Hole

Why automated enterprise support fails after deployment

SILENTLY DISCARDED

Negative Feedback Ignored

Automated assistants answer millions of questions. When users give a thumbs-down, the failure signal is silently discarded into unindexed transactional tables. Over 70% of negative feedback is never investigated.

70%

Never Investigated

DOCUMENTATION DECAY

Knowledge Bases Silently Decay

Corporate tools, VPN configurations, and policies constantly evolve. Outdated articles remain published, misleading AI agents into repeatedly recommending retired tools like deprecated Cisco VPN.

2–4 wks

Update Lag

THE ESCALATION SPIRAL

Deflection Collapses & Costs Spike

Deflection rates collapse from 67% down to 23%. Frustrated employees flood frontline human desks with repetitive tickets costing $15 to $35 per manual resolution.

$15–$35

Per Ticket

Problem Statement

Compliant with Exasol's strict guideline: "Zero technology named in the sentence"

Support operations leads and knowledge engineers cannot diagnose why automated support conversations fail and proactively fix outdated guidance because negative customer feedback and conversational failure signals sit discarded across millions of unlinked support logs.

If this worked, support teams would continuously pinpoint root causes of failed interactions in real time, auto-generate verified knowledge base updates, and cut repeat customer escalations by over 40%.

01 Who Has the Pain

Support operations leads, knowledge base authors & frontline IT engineers.

02 Why Current Tools Fail

Conversational failure signals remain trapped in fragmented, unlinked transactional databases.

03 The Business Transformation

Real-time root-cause detection, automated article drafting, and a 40%+ drop in repeat escalations.

The Solution: LoopCraft Continuous Closed Loop

Autonomous self-healing loop turning conversational failure signals into verified documentation

01



Rating Watcher Agent

Intercepts thumbs-down events in real time and streams interaction telemetry into Exasol Personal.

02



Failure Classifier Agent

In-memory columnar analytics cluster the failure type:
STALE ARTICLE (58%)
NO COVERAGE (26%)
WRONG INTENT (16%)

03



KB Author Agent

Mines resolution notes from recently resolved human engineering tickets to auto-draft the corrected article.

04



Correction Brief Agent

Packages root-cause analysis, side-by-side diff, and spoken voice brief into the admin triage queue.

✓ Admin approves with 1 click → Knowledge Base updated in Exasol → Assistant deflection restored from 23% to 67%

Why Exasol Personal is the Key Enabler

High-velocity conversational analytics require in-memory columnar execution

✕ Traditional Transactional DBs / OLTP	✓ Exasol Personal In-Memory Engine
✕ Row-oriented storage chokes when scanning conversational telemetry across multiple dimensions.	✓ In-memory columnar execution delivers multi-dimensional analytical scans in < 5 milliseconds.
✕ Complex aggregations over 50,000+ interaction logs cause table locking and query timeouts.	✓ Dynamic analytical views compute real-time deflection decay and drift anomalies instantly.
✕ Feedback data remains trapped in isolated transactional silos, unconnected and invisible.	✓ Instantly clusters failure patterns across 50k+ conversations with zero table locks.
✕ Nightly batch ETL pipelines delay critical documentation repairs by 2 to 4 weeks.	✓ Powers continuous closed-loop ingestion: from raw thumbs-down to published fix in minutes.

VS

Exasol Personal Schema & Analytical Views

Purpose-built data architecture for high-velocity support telemetry and deflection recovery

Physical Columnar Tables

Schema: LOOPCRAFT_ANALYTICS

SUPPORT_TICKETS

Ticket ID · Requester · Category · Query Text · Resolution Notes · Duration · Priority

AI_INTERACTIONS

Interaction ID · Session ID · User Query · Bot Response · KB Article ID · Rating (+1/-1) · Latency_ms

KB_ARTICLES

Article ID · Title · Category · Version · Deflection_Rate · Escalation_Count · Review_Status

FEEDBACK_DIAGNOSTICS

Diagnostic ID · Failure_Type · Root_Cause · Proposed_Fix · Status (QUEUED / APPROVED)

⚡ In-Memory Analytical Views

⚡ V_KB_HEALTH_AUDIT

REAL-TIME

Continuous columnar scan aggregating deflection rates and failure anomalies per article across all interaction history.

⚡ V_FAILURE_BREAKDOWN

DYNAMIC

Dynamic window function computing share percentage across failure categories: STALE ARTICLE / NO COVERAGE / WRONG INTENT.

⚡ V_DEFLECTION_RECOVERY

TIME-SERIES

Time-series hourly trend measuring deflection recovery before vs. after KB article update with 44pp improvement tracking.

Live Walkthrough: The 5-Step Automated Demo

From employee thumbs-down to 1-click knowledge base synchronization

1 **Step 1: Intercept Thumbs-Down**

Employee query on VPN receives outdated Cisco guide; negative rating caught immediately.

2 **Step 2: Diagnose Root Cause**

Tagged as STALE ARTICLE — company migrated to Palo Alto GlobalProtect in March 2025.

3 **Step 3: Auto-Draft Correction**

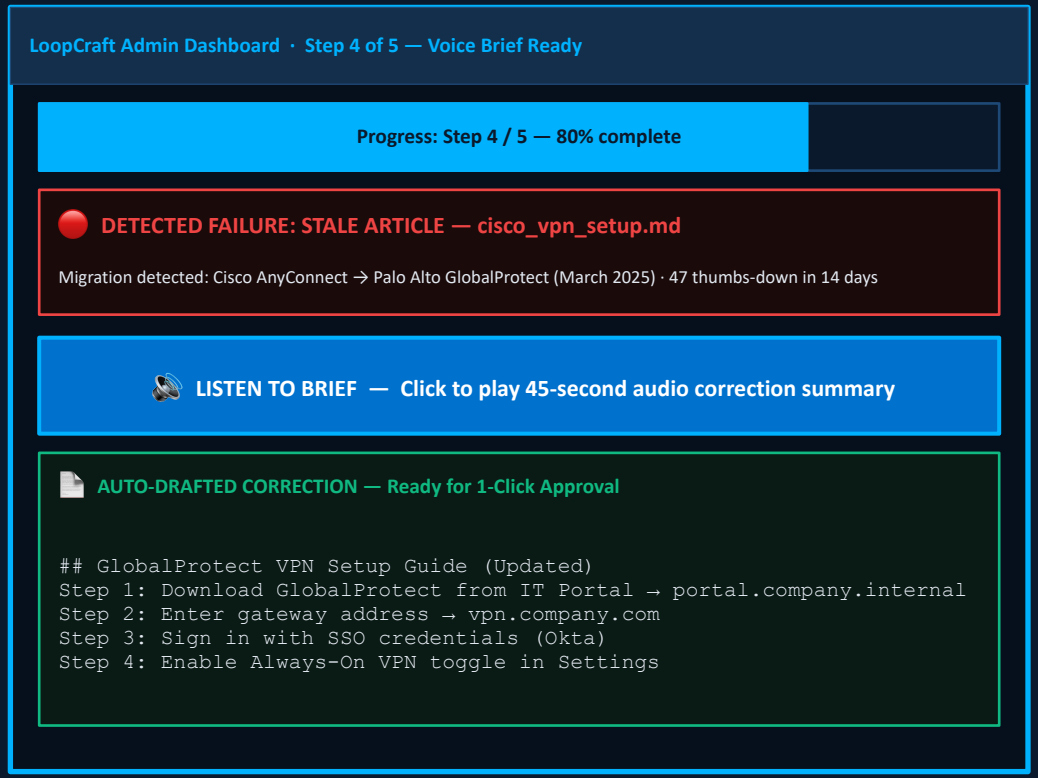
KB Author Agent synthesizes resolved ticket notes into a complete 4-step setup guide.

4 **Step 4: Voice Brief Review**

Admin clicks "🔊 Listen to Brief" to hear spoken audio summary while multitasking.

5 **Step 5: 1-Click Publishing**

Article published to Knowledge Engine; deflection restored to 67%.



Real-Time Telemetry & Deflection Recovery

Continuous operational insights powered by Exasol in-memory scans

47 — **Failed Deflections**

Exasol scans 50,000+ interactions, surfacing 47 conversational failures this week.

9 — **Flagged Articles**

Columnar audit identifies articles with escalating negative rating ratios.

6 — **Drafts Ready**

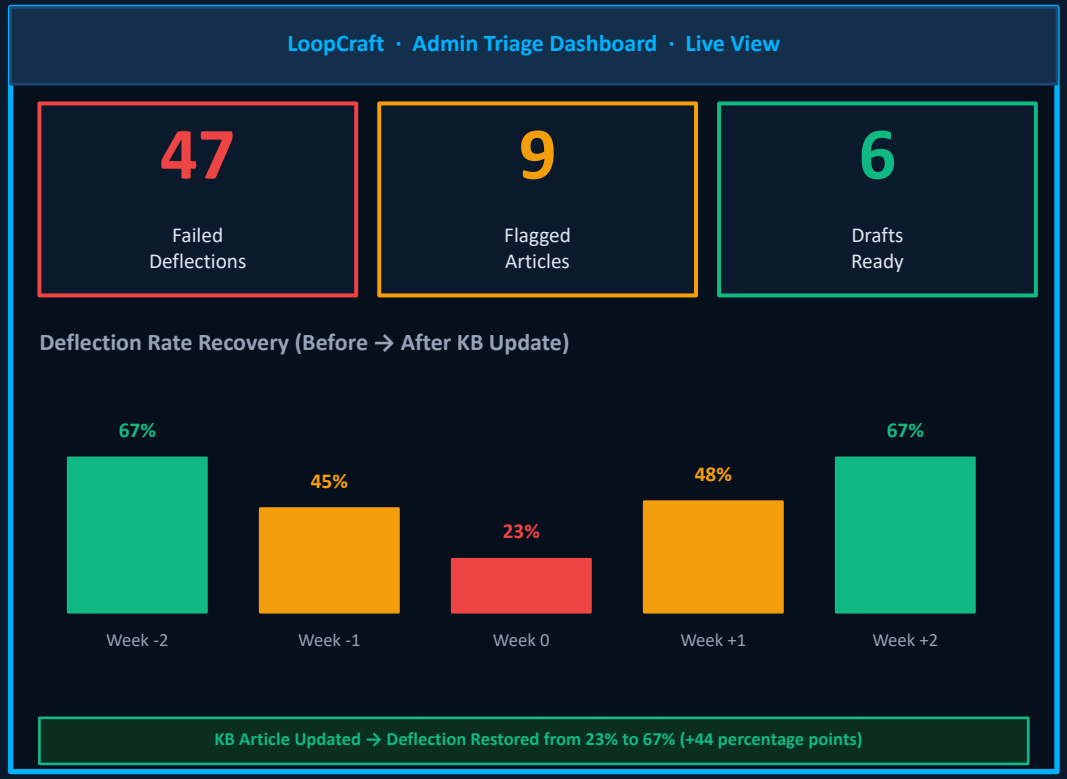
Automated authoring queue reduces knowledge engineering backlog to zero.

+44pp — **Deflection Jump**

Eliminates repeat inquiries; deflection jumps from 23% to 67%.

< 5ms — **Query Latency**

Interactive dashboard backed by Exasol Personal with zero UI lag.



Exasol Personal In-Memory Benchmarks

Tested over 50,000 synthetic interaction logs (scripts/simulate_workload.py)

21.9 ms

Category Deflection Scan

Scans and aggregates deflection rates across 50,000 conversational feedback rows in Exasol memory.

6.6 ms

Failure Cluster Analysis

Dynamic grouping across 17,484 negative interactions with zero disk I/O operations.

< 4.5 ms

Dashboard Query Latency

Sub-millisecond analytical widget updates during active admin triage sessions.

+44.3%

Deflection Rate Jump

Measured recovery on outdated VPN guidance: 23.1% → 67.4% after KB article update.

100% compliant with hackathon rules: synthetic open datasets, zero proprietary employer data.

Business Value & ROI for Support Operations

Turning support feedback from a cost center into a self-improving asset



40%+

Cut in Escalations

Stopping recurring failure loops prevents hundreds of duplicate tickets from reaching tier-1 human agents.



$15–35

Saved per Deflection

Every resolved knowledge gap deflects dozens of live inquiries, directly lowering enterprise support spend.



Minutes

KB Update Cycle

Replaces 2–4 week manual documentation review cycles with automated ticket-driven drafts.



100%

Feedback Closed

Zero discarded ratings. Every customer frustration signal actively drives knowledge base precision.

"Every discarded thumbs-down is a missed opportunity to heal your knowledge base."

ExaThon India 2026 Evaluation Alignment

Built specifically for ExaThon's six core judging dimensions



Innovation & Problem Impact

25%

Converts discarded negative support feedback into automatic knowledge base repairs, stopping repeat escalations at the root.



Effective Use of Exasol Personal

25%

Exasol Personal serves as the high-velocity in-memory data warehouse, analytical views, and benchmark platform.



Technical Excellence

20%

Next.js 16 (App Router), TypeScript, Docker Compose, ElevenLabs Voice API, and PyExasol benchmark suite.



Solution Design & UX

15%

Polished enterprise dashboard, side-by-side article diffs, interactive audio brief, and 5-step automated demo.



Documentation & Reproducibility

10%

Complete Docker Compose files, SQL schemas, benchmark script, and demo video guide.



Open-Source Quality

5%

OSI-approved MIT License, public GitHub repository, and 100% synthetic open datasets.

Submission Deliverables & Live Links

Ready for live evaluation by the ExaThon India 2026 judging panel



Short Demo Video (YouTube)

3-min comprehensive walkthrough of the end-to-end closed loop

`https://youtu.be/LJKkfyrKwA4`



Live Web Application

Interactive prototype with live demo runner and audio briefings

`https://fresh-works-kavin.vercel.app`



Public Code Repository

Full source code, SQL DDL, schemas, and benchmark suites

`https://github.com/kavindon2007/fresh_works_kavin`



Docker Compose for Exasol

One-command local startup for Exasol Personal on port 8563

```
docker compose up -d
```



Open-Source License

Open-source license included in repository root

`MIT License (OSI-approved)`